

IPA ASSIGNMENT - 23

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1. Image compression using erosion technique using a black white image

```
import numpy as np
import matplotlib.pyplot as plt
from PIL import Image
def erosion(image, kernel):    h, w = image.shape
    kh, kw = kernel.shape    ph, pw = kh // 2, kw // 2
    padded=np.pad(image, ((ph, ph), (pw, pw)), mode="constant", constant_values=0)
    output = np.zeros_like(image)
    for i in range(h):    for j in range(w):
        region = padded[i : i + kh, j : j + kw]
        if np.all(region[kernel == 1] == 255):            output[i, j] = 255
    return output
image = Image.open("img_2.jpeg").convert("L") image = np.array(image)
binary = np.where(image > 127, 255, 0).astype(np.uint8)
kernel = np.ones((3, 3), dtype=np.uint8) result = erosion(binary, kernel)
plt.figure(figsize=(10, 5)) plt.subplot(1, 2, 1)
plt.imshow(binary, cmap="gray") plt.title("Original Binary Image")
plt.axis("off") plt.subplot(1, 2, 2)
plt.imshow(result, cmap="gray") plt.title("Eroded Image")
plt.axis("off") plt.tight_layout() plt.show()
Image.fromarray(result).save("eroded_image.png")
```

OUTPUT

